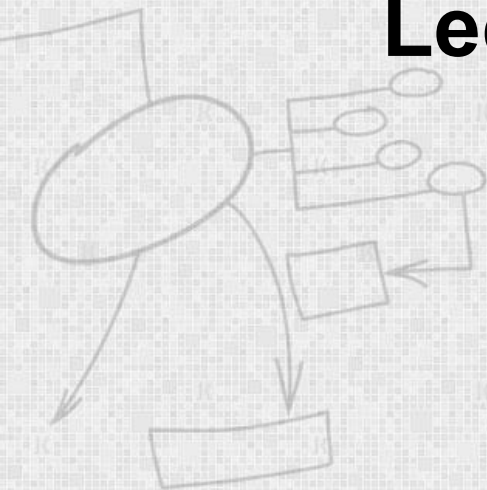


Database Engineering

Lecture 3: The Enhanced ER Model

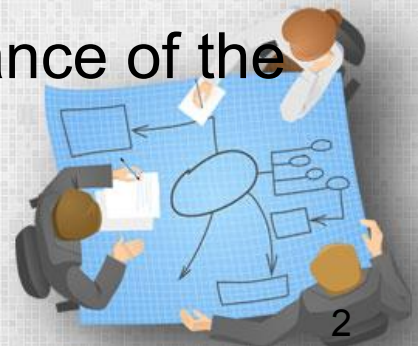
By

Dr Wasi Haider Butt



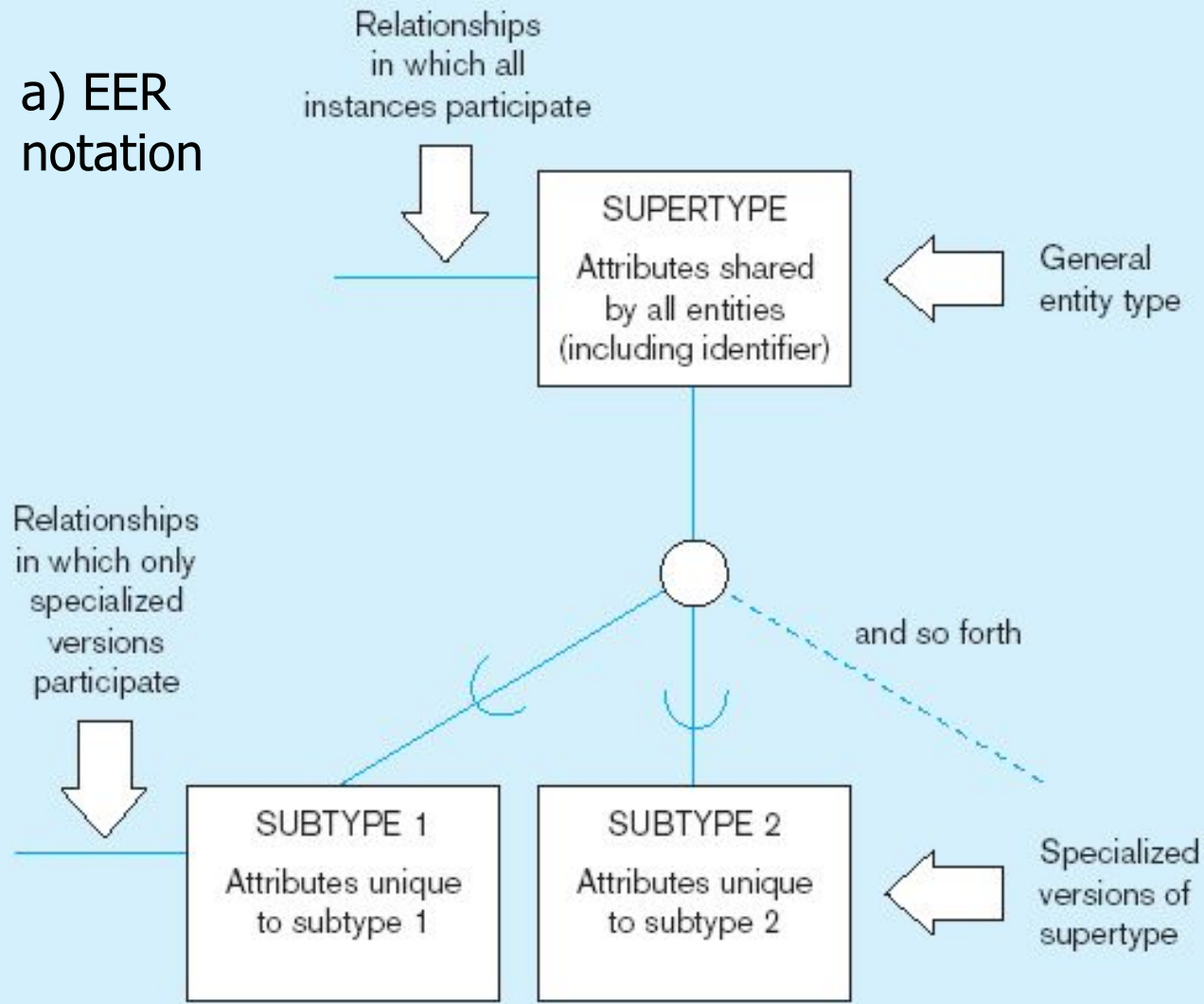
Supertypes and Subtypes

- **Subtype:** A subgrouping of the entities in an entity type that has attributes distinct from those in other subgroupings
- **Supertype:** A generic entity type that has a relationship with one or more subtypes
- **Attribute Inheritance:**
 - Subtype entities inherit values of all attributes of the supertype
 - An instance of a subtype is also an instance of the supertype



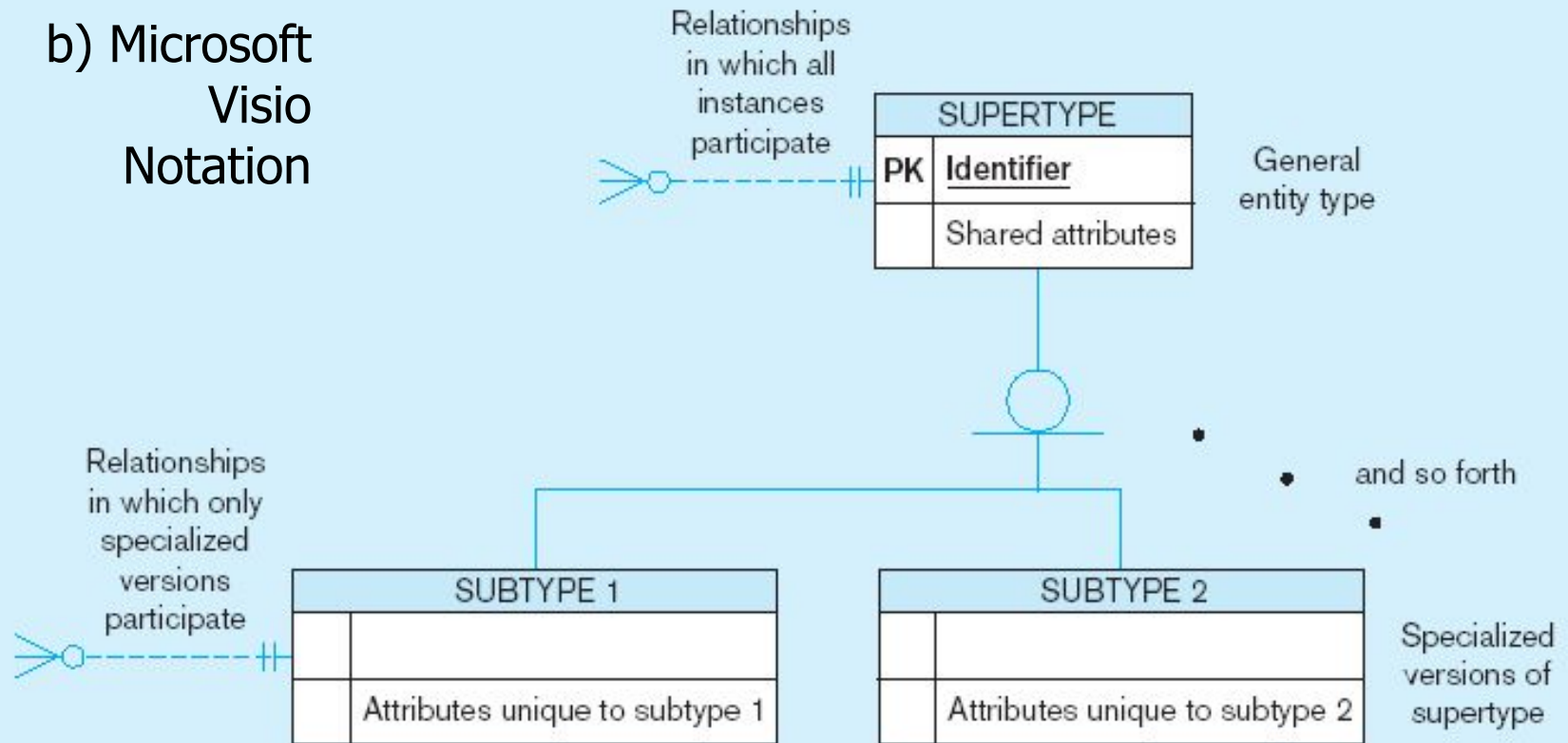
Basic notation for supertype/subtype notation

a) EER notation



Basic notation for supertype/subtype notation (cont.)

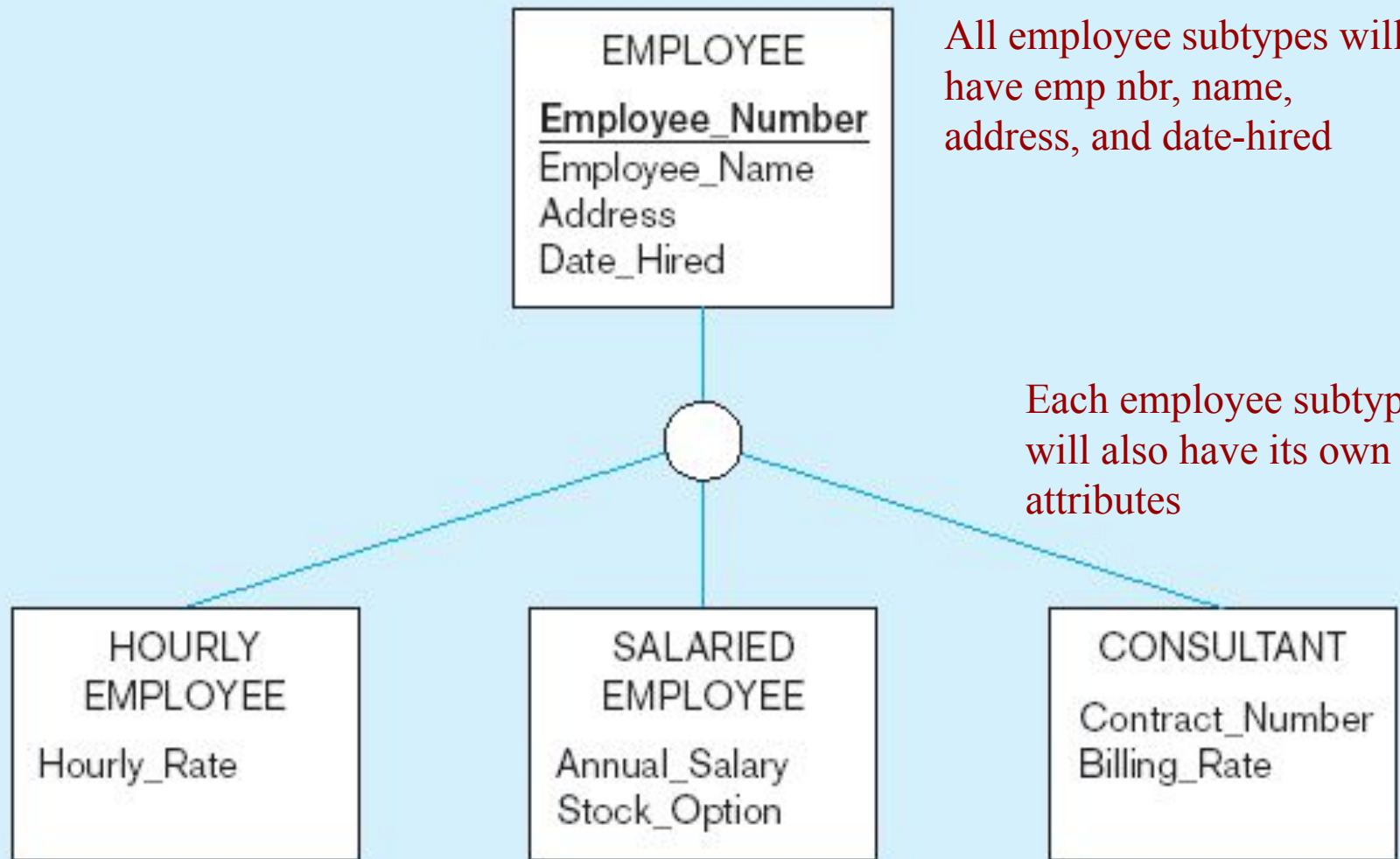
b) Microsoft Visio Notation



Different modeling tools may have different notation for the same modeling constructs



Employee supertype with three subtypes

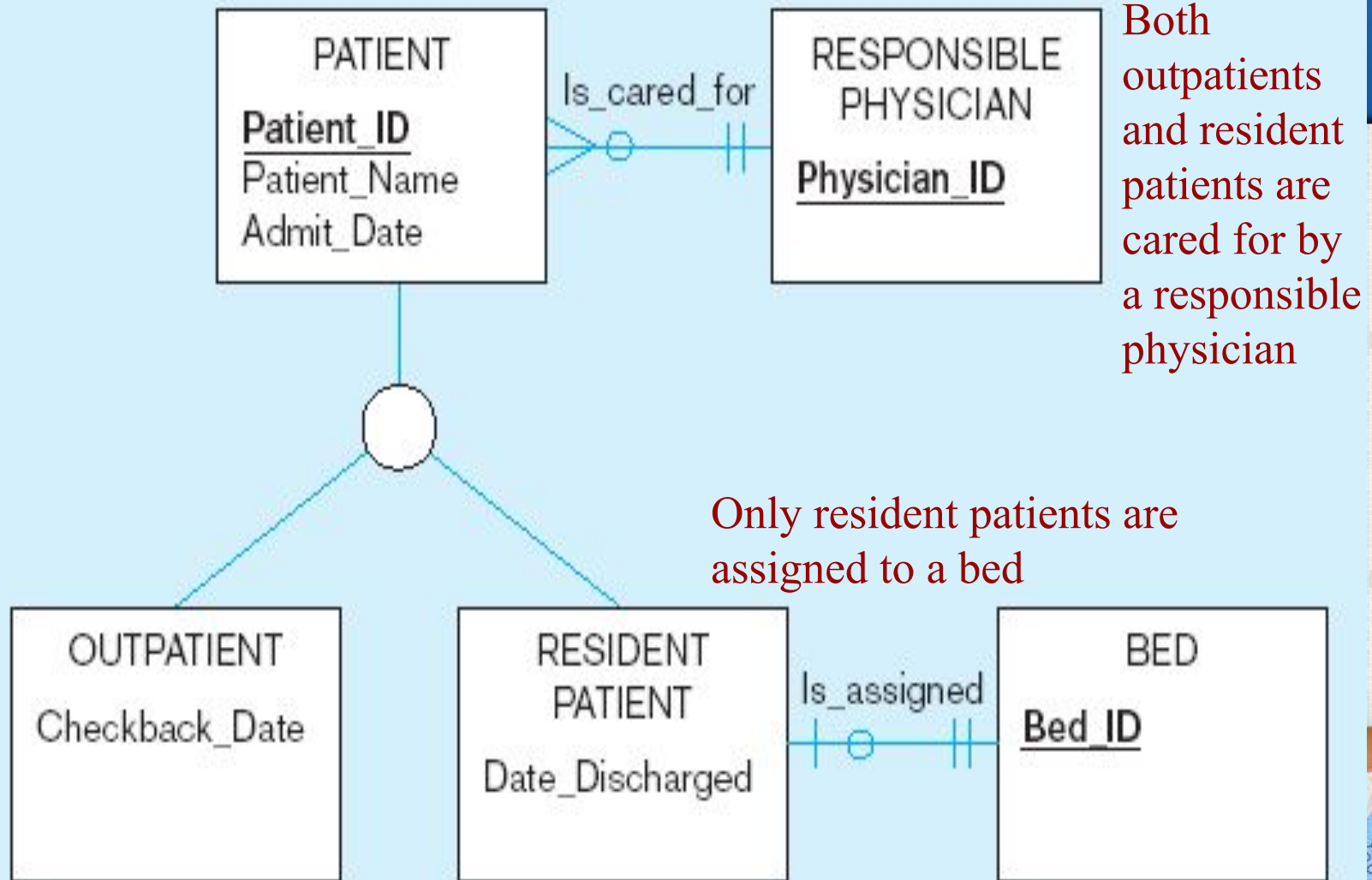


Relationships and Subtypes

- Relationships at the **supertype** level indicate that all subtypes will participate in the relationship
- The instances of a **subtype** may participate in a relationship unique to that subtype. In this situation, the relationship is shown at the subtype level



Supertype/subtype relationships in a hospital



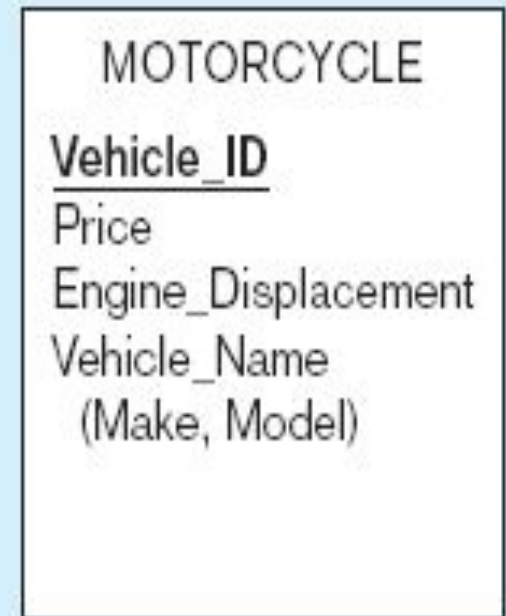
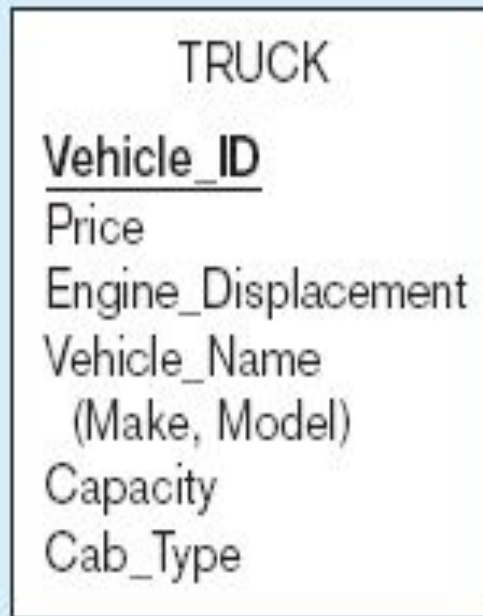
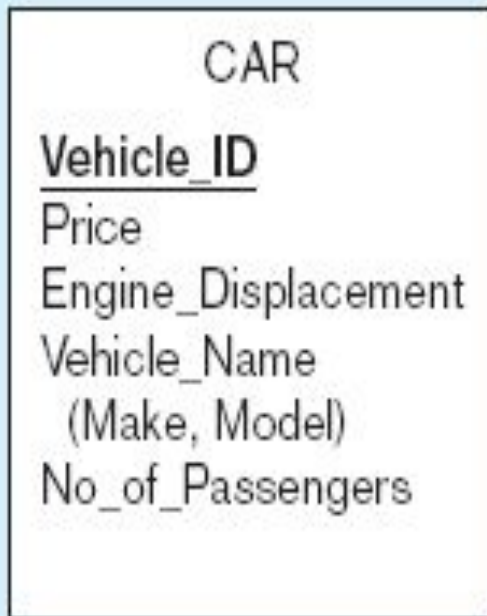
Generalization and Specialization

- **Generalization:** The process of defining a more general entity type from a set of more specialized entity types.
BOTTOM-UP
- **Specialization:** The process of defining one or more subtypes of the supertype and forming supertype/subtype relationships. TOP-DOWN



Example of generalization

a) Three entity types: CAR, TRUCK, and MOTORCYCLE

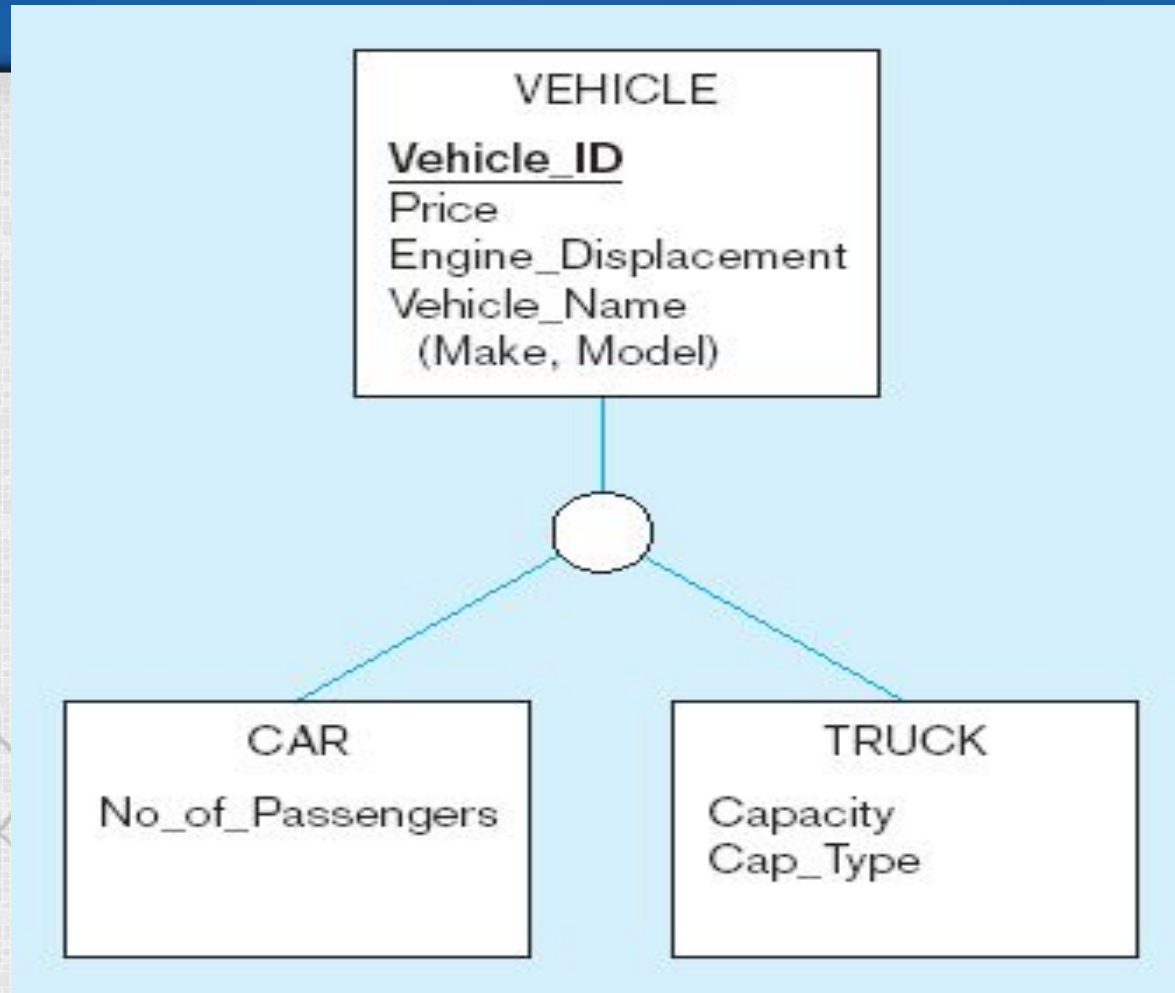


All these types of vehicles have common attributes



Example of generalization (cont.)

b) Generalization to VEHICLE supertype



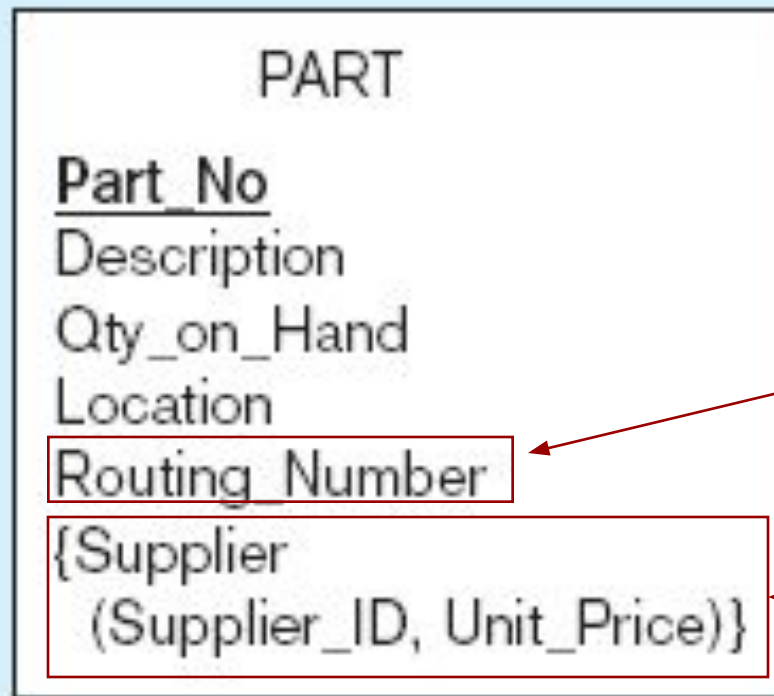
So we put
the shared
attributes in
a supertype

Note: no subtype for motorcycle, since it has no unique attributes



Example of specialization

a) Entity type PART



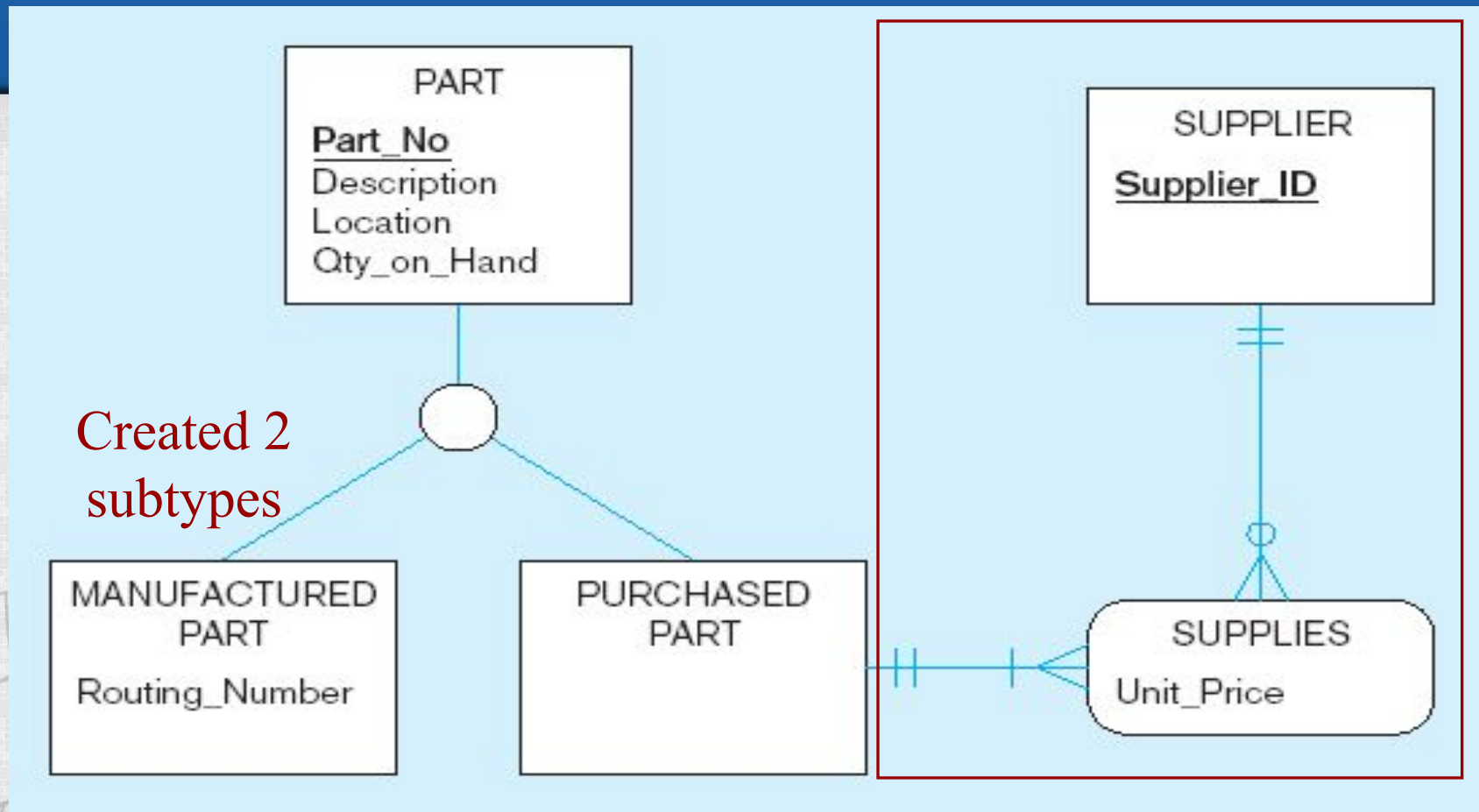
Only applies to manufactured parts

Applies only to purchased parts



Example of specialization (cont.)

b) Specialization to MANUFACTURED PART and PURCHASED PART



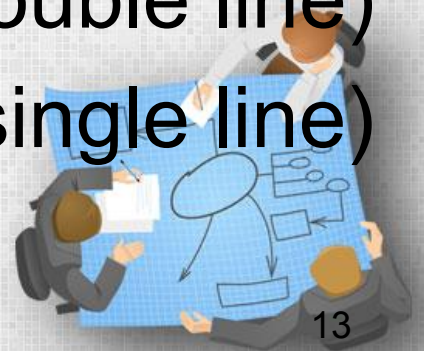
Note: multivalued attribute was replaced by an associative entity relationship to another entity

Constraints in Supertype/ Completeness Constraint

- **Completeness Constraints:**

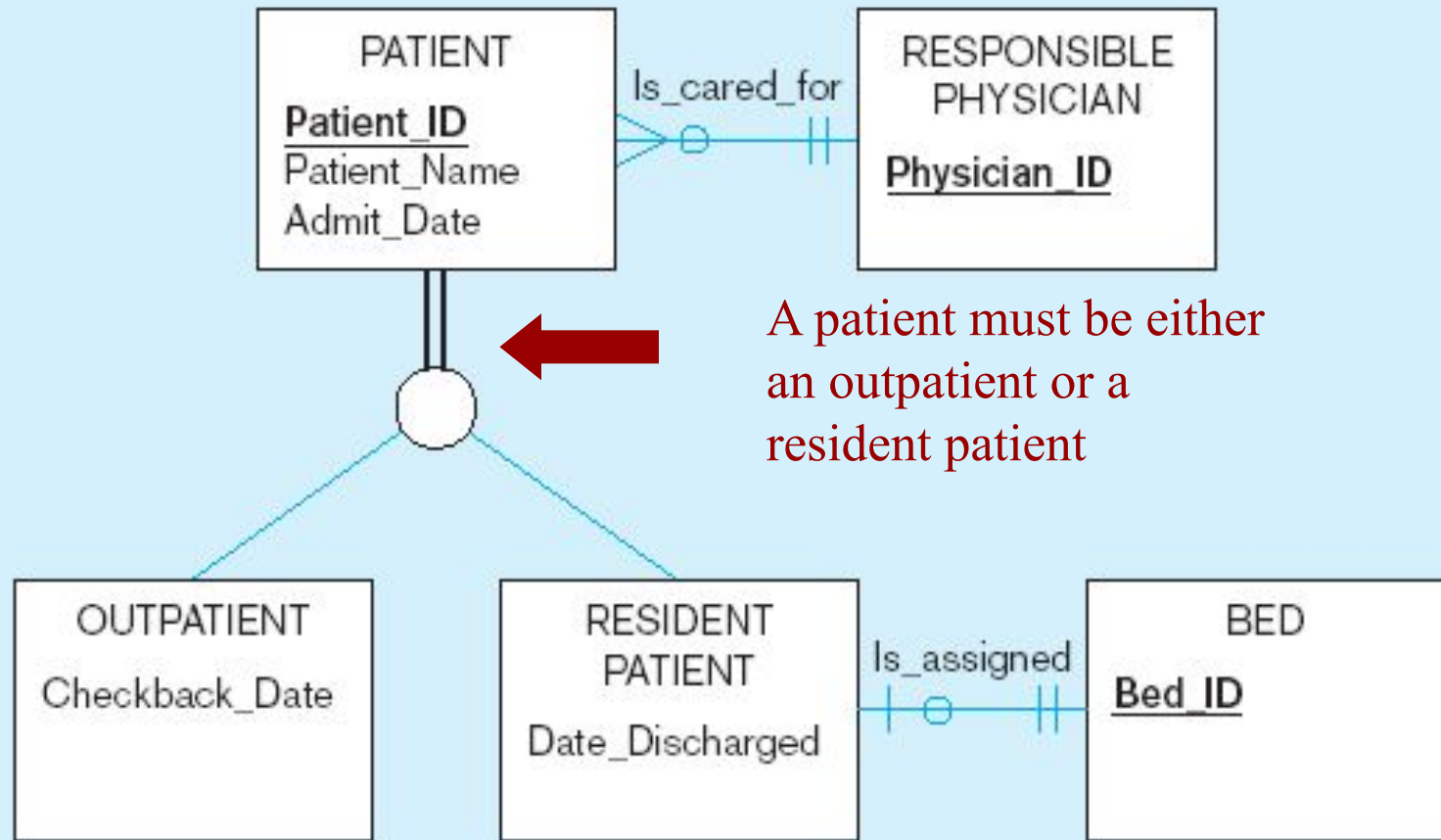
Whether an instance of a supertype ***must*** also be a member of at least one subtype

- Total Specialization Rule: Yes (double line)
- Partial Specialization Rule: No (single line)



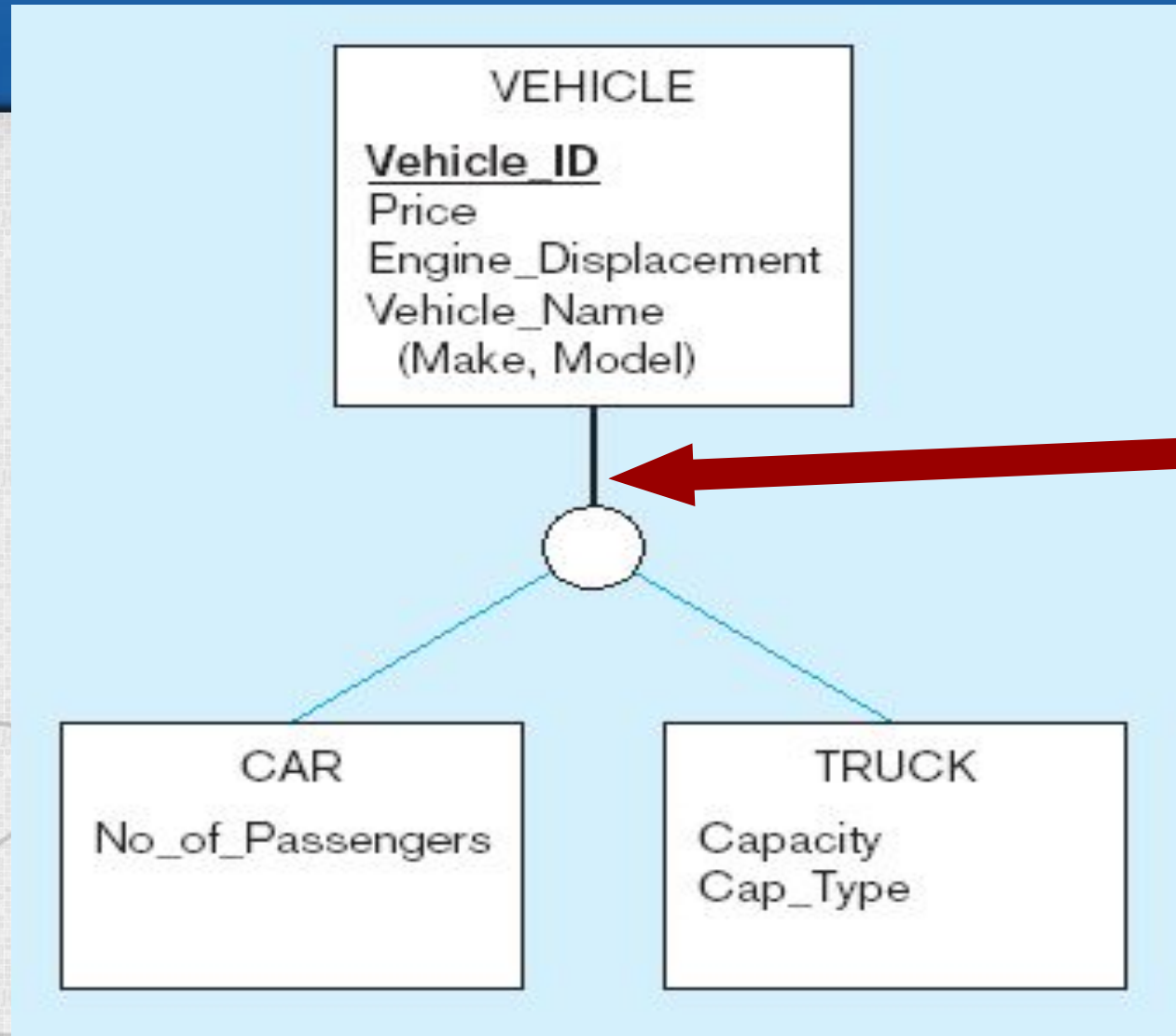
Examples of completeness constraints

a) Total specialization rule



Examples of completeness constraints (cont.)

b) Partial specialization rule

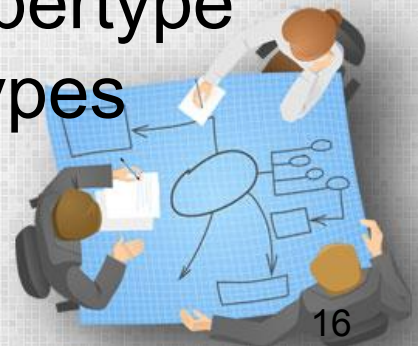


A vehicle
could be a
car, a truck,
or neither



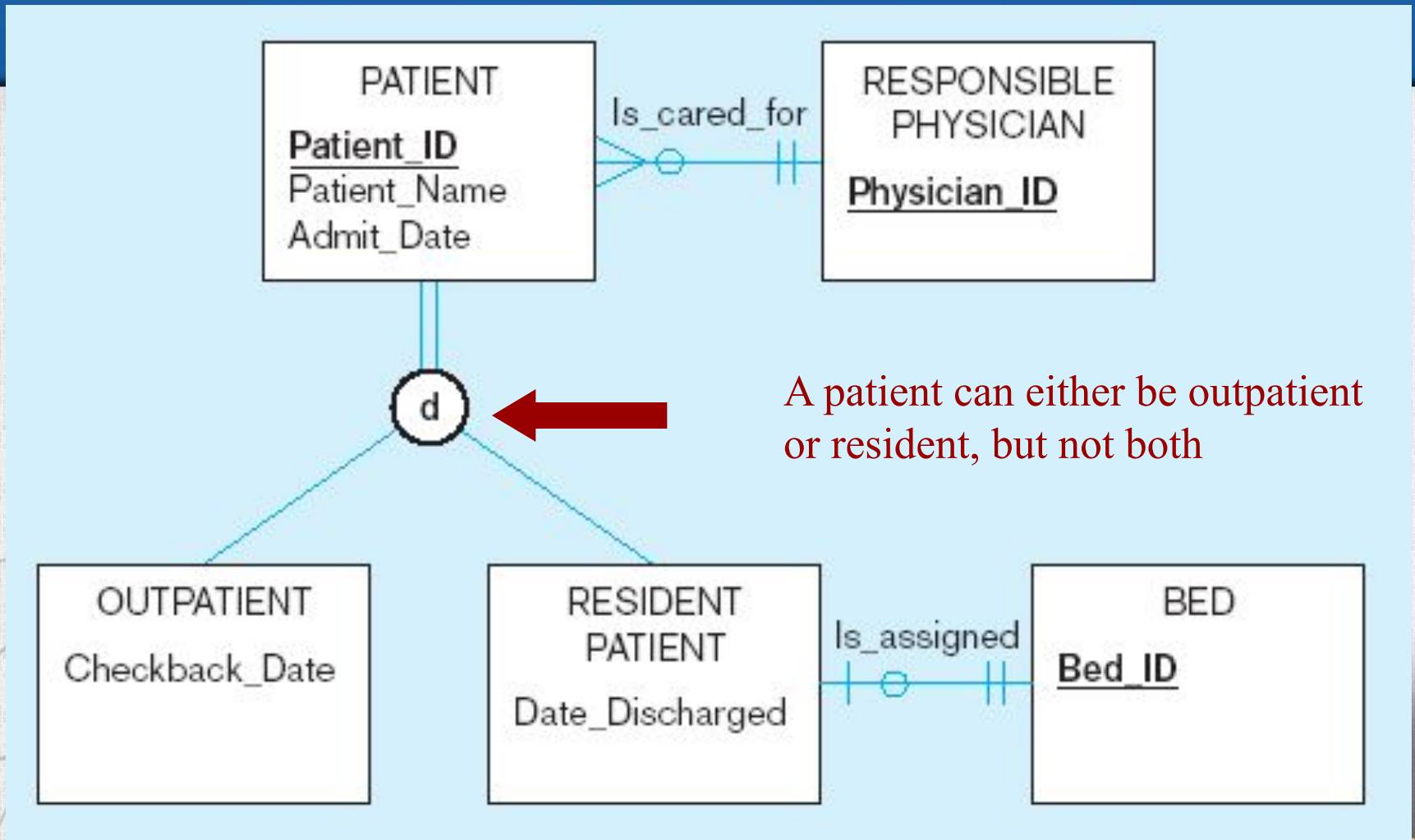
Constraints in Supertype/Disjointness constraint

- **Disjointness Constraints**: Whether an instance of a supertype may *simultaneously* be a member of two (or more) subtypes
 - Disjoint Rule: An instance of the supertype can be only ONE of the subtypes
 - Overlap Rule: An instance of the supertype could be more than one of the subtypes



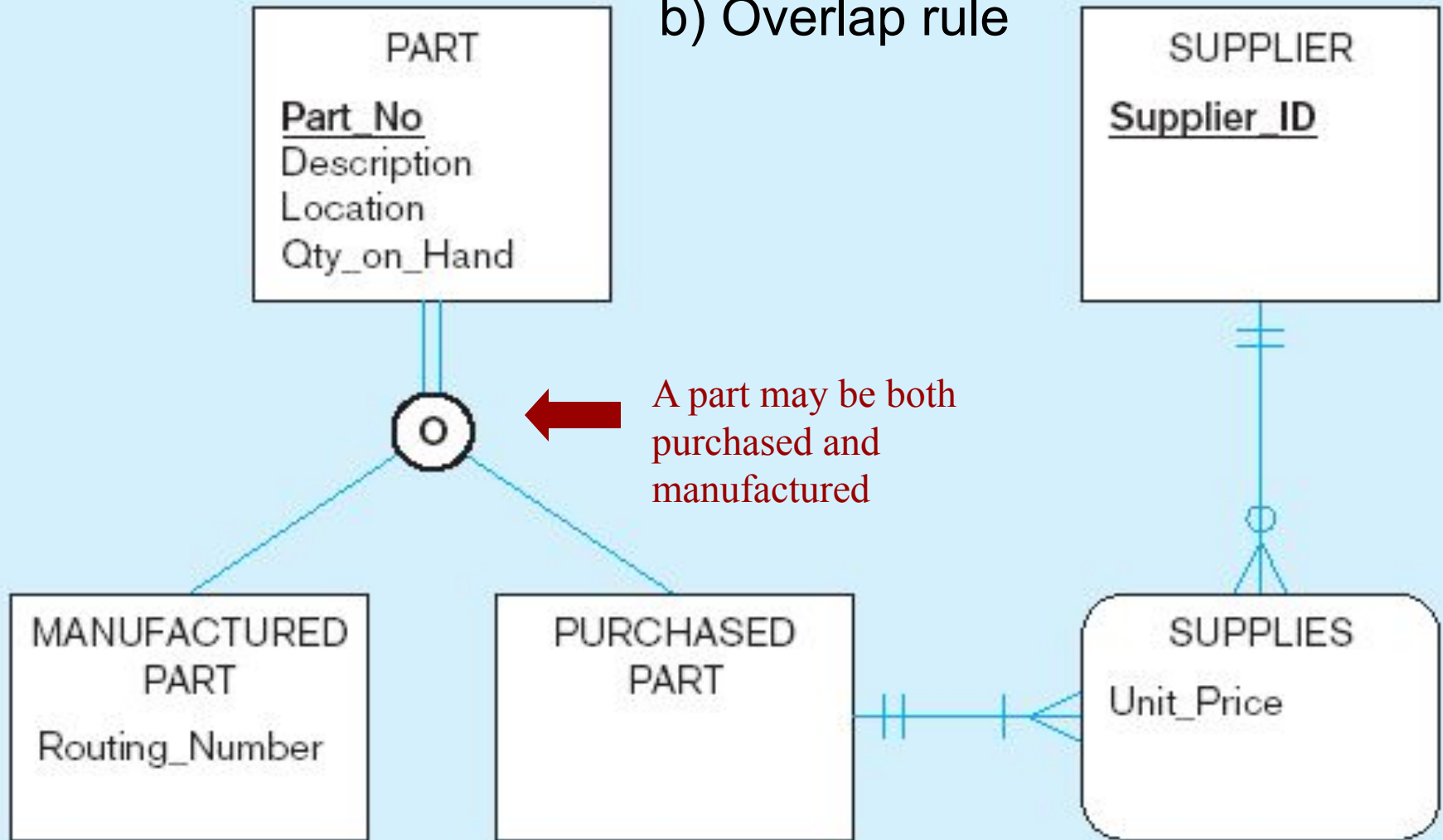
Examples of disjointness constraints

a) Disjoint rule



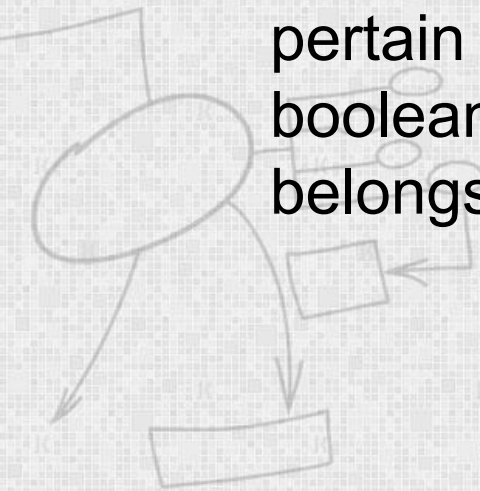
Examples of disjointness constraints (cont.)

b) Overlap rule

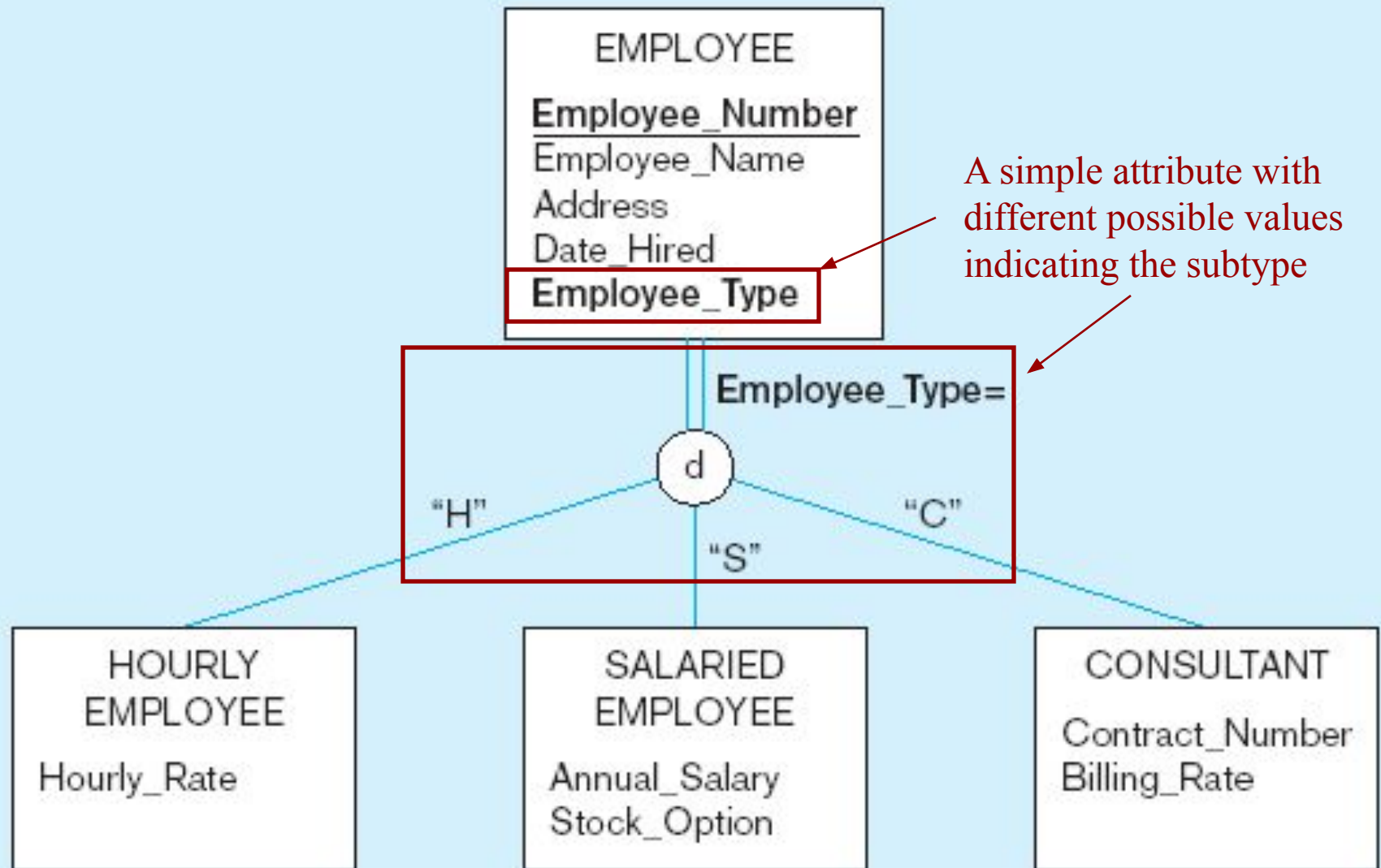


Constraints in Supertype/ Subtype Discriminators

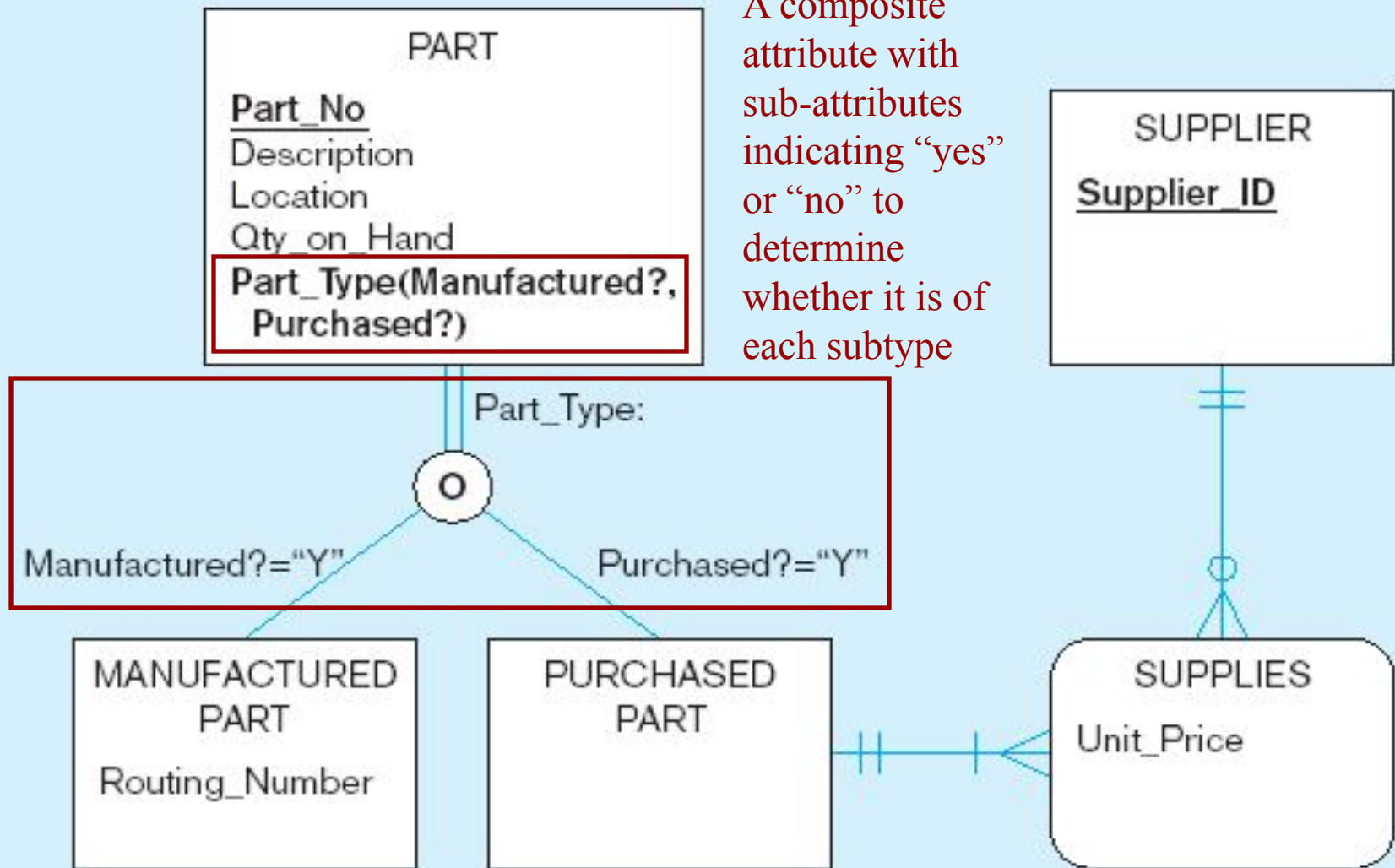
- **Subtype Discriminator**: An attribute of the supertype whose values determine the target subtype(s)
 - **Disjoint** – a *simple* attribute with alternative values to indicate the possible subtypes
 - **Overlapping** – a *composite* attribute whose subparts pertain to different subtypes. Each subpart contains a boolean value to indicate whether or not the instance belongs to the associated subtype



Introducing a subtype discriminator (*disjoint* rule)



Subtype discriminator (*overlap* rule)



A composite attribute with sub-attributes indicating “yes” or “no” to determine whether it is of each subtype



Example of supertype/subtype hierarchy

